

Mohanned Omer

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EDUCATION

Monash University

Bachelor of Computer Science, Data Science Specialisation

Melbourne, Australia

Grad: Dec 2027

British School Al Khubairat

A-Levels: A*A*A*A – UAE Golden Visa for Academic Excellence

- Student Award for dissertation titled: *To What Extent Is the UAE Equipped to Implement Face Recognition for Crowd Control?*

WORK EXPERIENCE

EY

Platforms Intern

Melbourne, Australia

Jul 2025 – Dec 2025

- Supported enterprise financial operations for a platform projected to serve **5M+ users** by configuring core components in **Dynamics 365 Finance**.
- Reduced manual testing time by **83%** (4 hours to 40 minutes) by automating validation workflows with **SQL** and tracking test results in **Azure DevOps**.
- Cut requirements documentation time by **60%** (8 to 3 hours per document) by building an AI-assisted document generation workflow using **Power Automate**.

Monash University

Computer Vision Research Intern

Melbourne, Australia

Dec 2025 – Mar 2026

- Built and stress-tested a real-time person re-identification system to maintain consistent identities across video frames by combining face and body embeddings, using **PyTorch**, **YOLOv8**, **DeepSORT**, and **CLIP-ReID**.
- Addressed limitations in detection-to-track matching by implementing **Hungarian matching** for globally optimal one-to-one assignment between new detections and existing person tracks.

PROJECTS

Supervisor Connect | Java, Spring Boot, TypeScript, Next.js, PostgreSQL, pgvector, OpenAI



- Built a full-stack research discovery platform with a **Spring Boot REST API** and **Next.js** frontend to index **480+** university research listings for semantic search using **OpenAI embeddings** and **pgvector**.
- Replaced rule-based categorisation with **Ward-linkage hierarchical clustering**; reduced filter latency from **3.3s to 1.5s** by consolidating joins, filtering, sorting, pagination, and counts into a single **PostgreSQL RPC**.

Mini PyTorch Framework & 3D U-Net | Python, Numba, CUDA



- Built a lightweight **PyTorch framework** from scratch, including neural-network modules and automatic differentiation.
- Developed parallel **CPU and GPU tensor backends** using **Numba** and **CUDA**, supporting map/reduce primitives, matrix multiplication, convolution, and pooling.
- Built and trained a **3D U-Net** for brain tumour segmentation, using patch-based training and sliding-window inference to process full MRI volumes within memory limits.

LEADERSHIP & COMMUNITY

Monash Data Science and AI Society

Data Insights Lead

Melbourne, Australia

Jun 2024 - Jun 2025

- Organized **6** technical workshops (**50+** attendees each) and delivered a Datathon with **250+** participants.
- Scaled the team from **15** to **45** members through recruiting and onboarding.
- Led an inter-university Datathon across **5** universities and delivered a data wrangling workshop to **200+** students.

TECHNICAL SKILLS

Programming Languages: Python, Java, R, SQL, Kotlin

Libraries/Frameworks: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, OpenCV

Tools/Platforms: Git, Tableau, Jupyter, Copilot Studio, Azure DevOps, Dynamics 365 Finance